

SET 6

PRODUCT SALES ANALYSIS

Dataset: Monthly Product Sales

Department	AIML
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
Name	Dudekula Mohammad Ilyas
Register Number	192325017
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Day	Thursday

Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

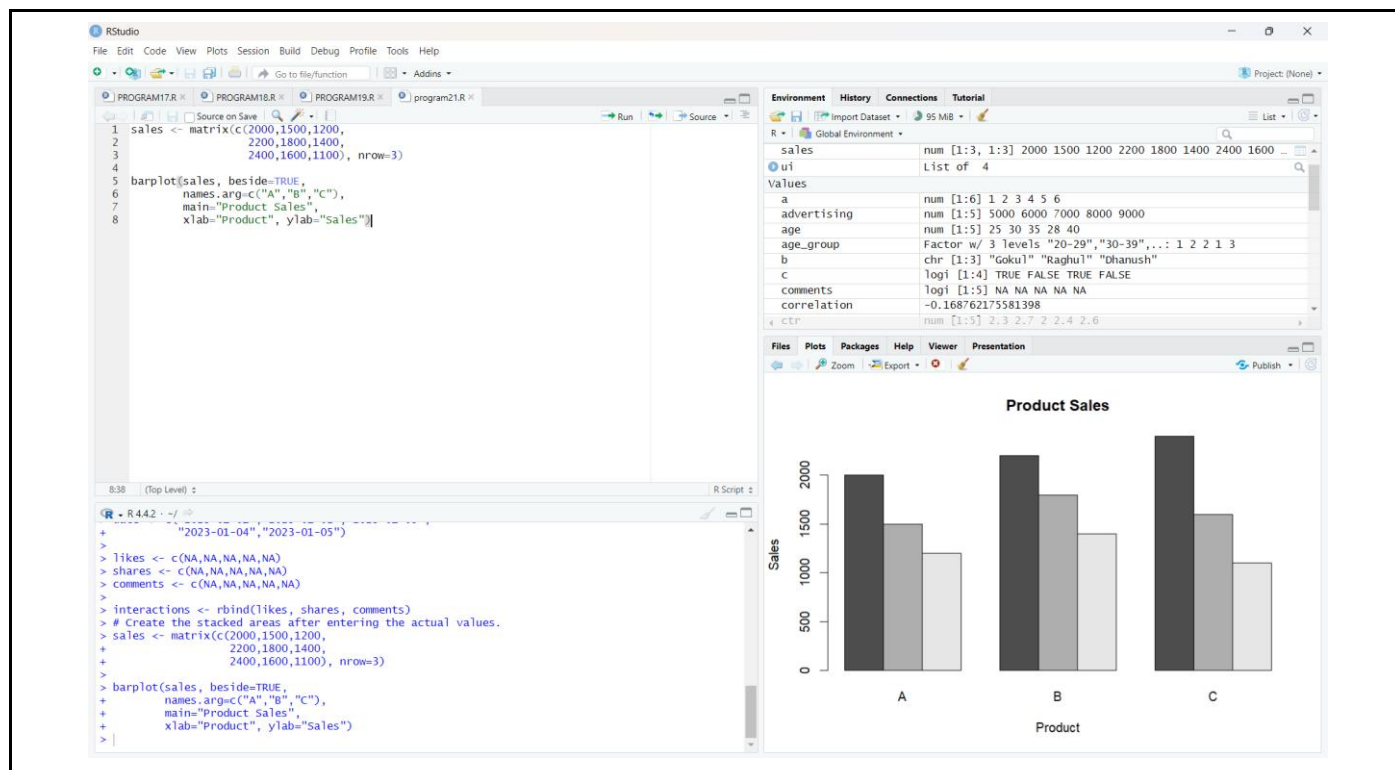
PROGRAM 1: Grouped Bar Chart

Aim: Create a grouped bar chart to visualize the sales of each product for the first quarter.

```
sales <- matrix(c(2000,1500,1200,
                  2200,1800,1400,
                  2400,1600,1100), nrow=3)

barplot(sales, beside=TRUE,
        names.arg=c("A","B","C"),
        main="Product Sales",
        xlab="Product", ylab="Sales")
```

Output:



Result: The grouped bar chart displays January, February, and March sales for Products A, B, and C.

PROGRAM 2: Stacked Area Chart

Aim: Generate a stacked area chart to represent the overall sales trend for all products over the three months.

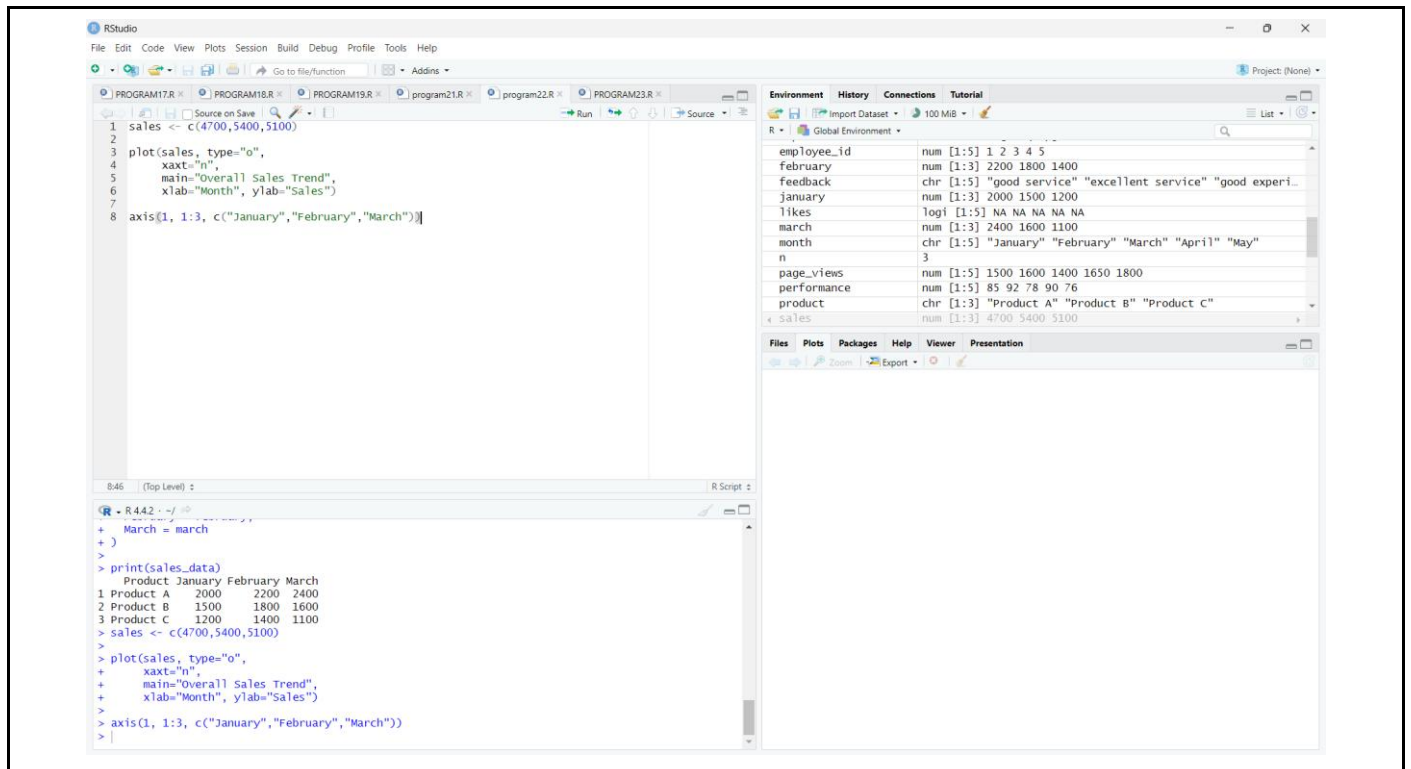
```
sales <- matrix(c(2000,1500,1200,
                  2200,1800,1400,
                  2400,1600,1100), nrow=3)

months <- c("January", "February", "March")
product_names <- c("Product A", "Product B", "Product C")

sales_ts <- rbind(sales[,1], sales[,2], sales[,3])
colnames(sales_ts) <- product_names
rownames(sales_ts) <- months

matplot(1:3, sales_ts, type="o", lty=1,
        xlab="Month", ylab="Sales",
        main="Overall Sales Trend")
axis(1, 1:3, months)
```

Output:



Result: The monthly sales trend is represented across January, February, and March.

PROGRAM 3: Monthly Sales Table

Aim: Build a table showing the monthly sales data for each product.

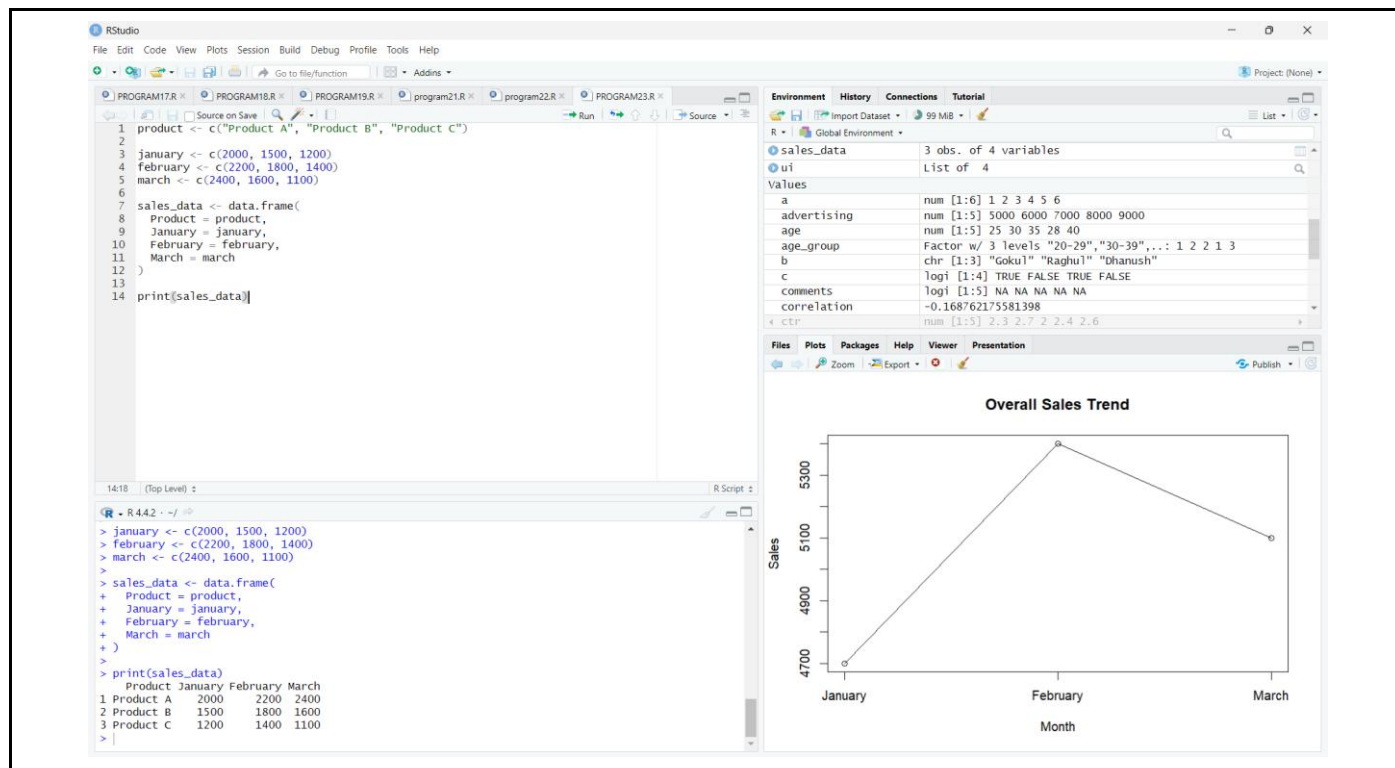
```
product <- c("Product A", "Product B", "Product C")

january <- c(2000, 1500, 1200)
february <- c(2200, 1800, 1400)
march <- c(2400, 1600, 1100)

sales_data <- data.frame(
  Product = product,
  January = january,
  February = february,
  March = march
)

print(sales_data)
```

Output:



Result: The table presents the January, February, and March sales values for each product.

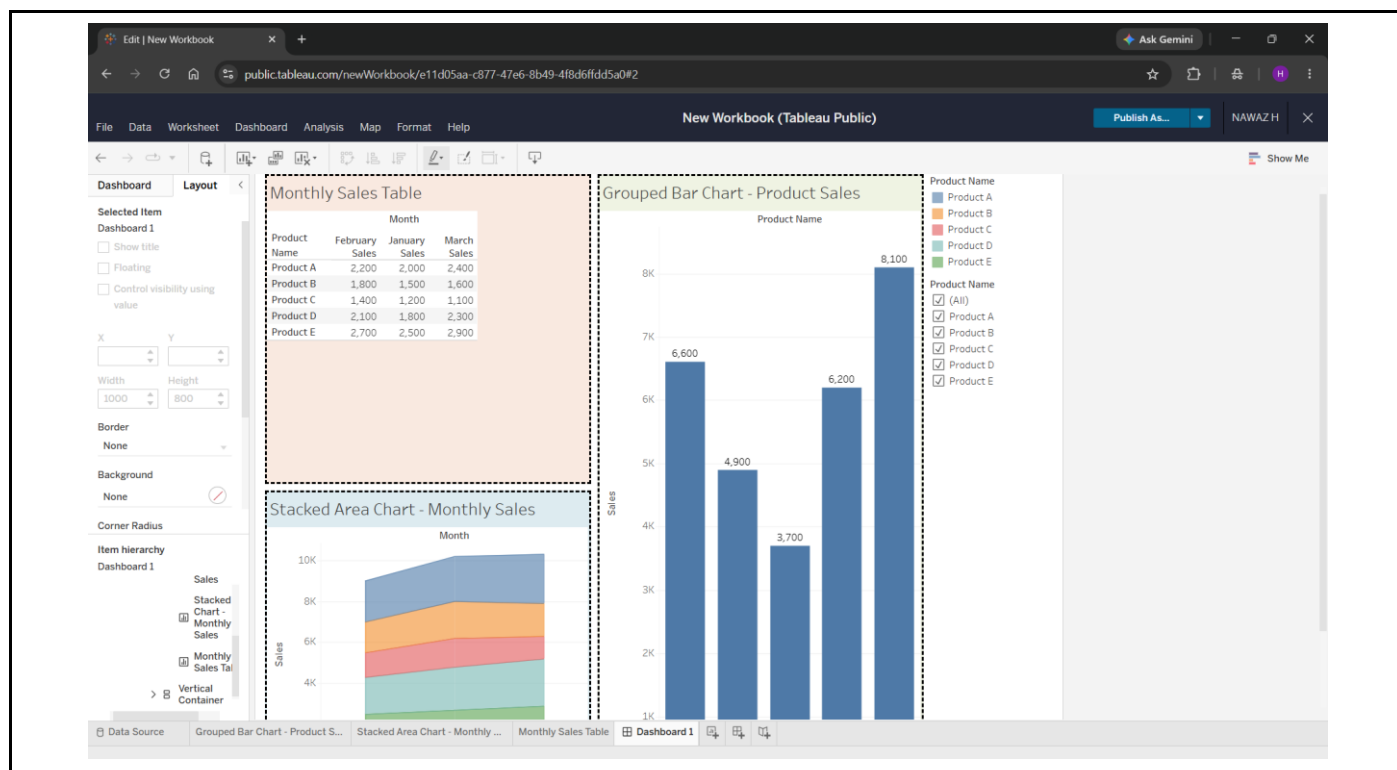
PROGRAM 4: Tableau Dashboard

Aim: Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and monthly sales table for interactive exploration of the sales data.

Tableau steps:

1. Connect the Monthly Product Sales dataset.
2. Create a grouped bar chart using Product Name and Sales.
3. Create a stacked area chart using Month, Sales, and Product Name.
4. Create a monthly sales table using Product Name and the three month sales fields.
5. Add all three sheets to a new dashboard.
6. Add the Product Name filter for interactive exploration.
7. Add titles and labels to the dashboard elements.

Output:



Result: The Tableau dashboard combines the required visualizations and table in one interactive dashboard.

END OF SET 6